

Rethinking LLM-Powered Devil’s Advocate Strategies for Supporting Minority Voices in Power-Imbalanced Group Decision-Making

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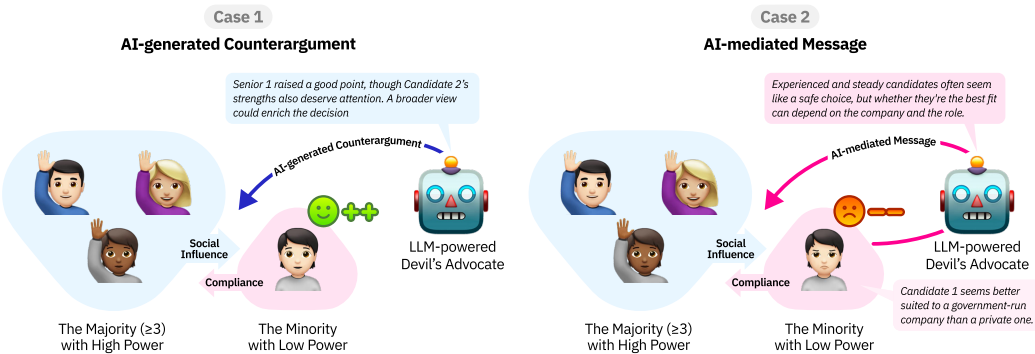


Fig. 1. LLM-powered Devil’s Advocate system mediates between majority and minority group members, presenting minority views through AI-generated counterargument to promote balanced group discussions.

Minority opinions are often suppressed in power-imbalanced group decision-making due to social pressure to comply with the majority. To address this problem, we developed an LLM-powered Devil’s Advocate system that aimed to foster attention to minority views through either AI-generated counterarguments or AI-mediated messages. We conducted a mixed-method experiment with 96 participants in 24 groups, comparing minority members’ experiences across baseline, AI-counterargument, and AI-mediated messaging conditions. Our findings revealed a nuanced trade-off: AI-generated counterarguments fostered a more flexible atmosphere and enhanced satisfaction, while AI-mediated messaging increased minority participation but unexpectedly reduced their psychological safety. This research contributes empirical evidence on how different AI implementations affect group dynamics, identifies a critical support paradox between participation and psychological safety, provides design implications for future systems, and highlights ethical challenges in implementing AI-mediated communication in hierarchical settings. These insights advance understanding of designing more equitable AI support for power-imbalanced group decision-making.

CCS Concepts: • **Human-centered computing** → **Computer supported cooperative work**; *Collaborative interaction*; *Natural language interfaces*; *HCI theory, concepts and models*.

Additional Key Words and Phrases: group decision-making, conversational agents, critical thinking, social influence, llm

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1 Introduction

Power imbalances in group decision-making frequently suppress minority perspectives, restrict the diversity of ideas, and weaken overall outcomes [50]. Compliance pressures often lead individuals to publicly align with the majority despite private disagreement, undermining psychological safety and discouraging meaningful participation from less-empowered members [54, 70]. While collaborative processes help groups solve complex challenges, support individual learning, and lead to more accurate and creative outcomes across domains such as healthcare, education, and research [36, 65, 87, 88], these benefits are diminished when hierarchy or conformity suppresses dissent. In an effort to address barriers to equal participation, recent studies have introduced AI-powered systems that facilitate idea generation, consensus-building, and alternative viewpoint introduction to encourage engagement and mitigate groupthink [14, 55, 82]. These early AI interventions show potential in improving group atmosphere and promoting more equitable discussions.

However, a significant research gap exists: current systems rarely provide direct support for minority opinions in contexts with strong power imbalances. Most AI interventions focus on group level dynamics or aggregate participation but fail to address the heightened risks, vulnerabilities, and compliance pressures marginalized members face when challenging the majority. Technical approaches such as anonymous feedback platforms and automated counterargument generators aim to diversify discussion but often fail to represent authentic minority views or adequately protect psychological safety and agency [23, 47, 58]. No current method guarantees that minority group members can express dissent without risking exposure, backlash, or social consequences, particularly in hierarchical settings where anonymity proves difficult to maintain and dissent carries professional or social costs. Consequently, group decision making under power imbalances continues to overlook critical minority perspectives, and current AI approaches remain limited in creating psychologically safe, equitable spaces for all participants.

To address this gap, this research explores how an LLM-powered Devil’s Advocate system can improve group decision-making by supporting minority voices in power-imbalanced settings. Our system uniquely serves as both an independent contributor of alternative perspectives and a surrogate that can anonymously voice minority opinions. This dual functionality allows minority members to express dissenting views without fear of social consequences, potentially enabling individuals experiencing compliance pressure to feel more psychologically secure, participate more actively, and gain greater satisfaction with group decision processes. We investigate four research questions focused on participants’ experiences:

- **RQ1.** In what ways does an LLM-powered Devil’s Advocate system influence minority members’ perceptions of psychological safety and marginalization in group decision-making?
- **RQ2.** How does the integration of such an AI system impact the extent and nature of minority participation and contribution patterns during group discussions?
- **RQ3.** What effect does the presence of an LLM-powered Devil’s Advocate have on members’ satisfaction with the decision-making process and the perceived quality of group outcomes?
- **RQ4.** How might using this AI system alter the cognitive demands placed on users navigating power-imbalanced discussions?

To investigate these questions, we conducted a mixed-methods study with 96 participants organized into 24 groups of four members each. Each group included three members assigned

as high-power majority (seniors) and one member as low-power minority (junior), with roles randomly assigned. We employed a mixed experimental design with participant type (majority vs. minority) as a between-subjects variable and type of LLM-powered Devil's Advocate as a partially within-subjects variable. Each participant first experienced a baseline discussion without AI involvement. In the second discussion, we divided the groups into two experimental conditions (AI-generated Counterargument (AIGC) Condition vs AI-mediated Message (AIMM) Condition). In one, an AI system took on a Devil's Advocate role, automatically generating counterarguments and alternative perspectives during AIGC condition. On the other hand, the minority group member could communicate privately with the AI. The system then paraphrased and presented these opinions as its own contributions, providing anonymity for the minority member's input during AIMM condition. Our findings indicate that AI-generated counterarguments from the Devil's Advocate improved group atmosphere and participant satisfaction. In contrast, enabling the minority member to submit anonymous opinions through AI increased discussion but reduced psychological safety and satisfaction for minority participants. These results underscore the complex trade-offs between amplifying minority voices and maintaining psychological safety when using AI-mediated communication in group decision-making.

This research makes four key contributions to CSCW. First, we empirically demonstrate how different LLM-powered Devil's Advocate implementations affect group dynamics under power imbalances, revealing that AI-generated counterarguments foster better discussions and satisfaction compared to AI-mediated minority messaging. Second, we identify a critical paradox in minority support: while AI-mediated messaging increases minority participation, it simultaneously decreases their psychological safety, challenging assumptions about anonymous communication channels. Third, we provide design implications for future AI-mediated group support systems, focusing on mental models, intervention timing, and transparent interfaces. Finally, we highlight ethical considerations for implementing AI systems in hierarchical decision-making contexts, particularly regarding potential power dynamic reinforcement. The remainder of this paper is structured as follows: we review related work on group decision-making and AI-mediated communication, describe methodology, present findings, discuss implications, and conclude with limitations and future research directions.

2 Related Work

2.1 The Impact of Social Influence and Power on Group Decision-making

Group decision-making leverages collective intelligence to produce superior outcomes across various domains [36, 65, 87], but these processes are significantly shaped by social influence and power dynamics [54, 70]. Social influence theory suggests that individuals tend to adjust their behavior to meet social demands, with the majority's opinions exerting particularly strong pressure on those with less power in the group. Moscovici's conversion theory specifically explains that multiple influences trigger a comparison process resulting in compliance - a form of conformity where individuals outwardly agree while maintaining private disagreement [70]. This compliance is typically direct, immediate, and temporary, serving as a coping mechanism in power-imbalanced situations rather than reflecting genuine belief change.

Power dynamics become especially problematic in hierarchical settings where power imbalances are formalized through reward and legitimate power structures [29]. Kelman's framework provides particular insight here, identifying compliance as an initial response to power where individuals conform primarily to avoid repercussions or gain rewards, rather than from genuine conviction [54]. This dynamic is especially evident among minority members, who are often treated as outgroup members and experience isolation. The effect is particularly pronounced when the size disparity

between majority and minority groups is substantial. The resulting self-censorship triggers a cascade of negative effects: as minority voices are silenced, groups lose access to diverse perspectives that could enhance decision quality, ultimately leading to groupthink, where the desire for consensus overrides critical evaluation of alternatives [49, 50, 53].

Traditional social psychology offers several approaches to address compliance and prevent its progression to groupthink in group settings. The devil's advocate technique addresses groupthink by assigning someone to challenge prevailing viewpoints, stimulating opinion diversity [64, 66, 72, 79, 80]. However, this method doesn't directly solve the underlying compliance issues and faces significant limitations: designated advocates often lack authentic dissenting perspectives, risk social ostracism when challenging powerful members, and cannot accurately represent unexpressed minority viewpoints [48, 72, 78]. Other compliance-reduction strategies include leadership interventions to create psychologically safe environments [25] and anonymous feedback channels [51]. However, these methods face practical challenges: leadership interventions rely on the leader's skills, and anonymous feedback often fails in small groups where unique opinions reveal identities. This research addresses these limitations by exploring how conversational AI might provide both psychological safety and true anonymity while preserving the benefits of diverse perspectives in decision-making processes.

2.2 AI-Enhanced Approaches to Improving Group Decision-Making

AI-assisted decision-making has evolved from individual-focused applications [8, 18, 57] to increasingly explore group contexts [63, 81, 96]. While researchers have developed various AI roles for group settings—including facilitators [55], creativity enhancers [41, 46], conflict mediators [22, 37], and consensus builders [74, 82, 86]—these approaches have revealed certain challenges. Groups may over-rely on AI recommendations without sufficient critical evaluation [13], and AI systems often struggle with the social nuances of group interactions [90, 97]. Despite these advances, existing research has critically overlooked how AI might specifically address power imbalances in group settings—situations where asymmetrical relationships allow certain members to exert disproportionate influence over discussions and decisions [28, 54].

This research gap around power imbalances frequently leads to the suppression of minority perspectives in group decision-making [50, 70]. The few studies that have attempted to support minority participants face a critical challenge: how to amplify marginalized voices without drawing unwanted attention to them. Researchers caution that overly targeted forms of support may inadvertently spotlight minority members, increasing discomfort and undermining the intended supportive function [23, 32, 33, 47]. Others caution that even well-intended interventions to support non-dominant members, such as less proficient speakers, can be perceived as disruptive or unfair, highlighting the complexity of fostering equitable participation in group interactions [58]. This research addresses this specific gap by investigating how LLM-powered Devil's Advocate systems can support minority perspectives in power-imbalanced groups without explicitly identifying these members. By examining AI mechanisms that challenge majority opinions while protecting vulnerable participants, we offer a novel approach to balancing power in group decision-making—one that promotes more equitable outcomes without compromising the psychological safety of minority voices.

2.3 AI-Mediated Communication for Group Interaction

AI-mediated communication (AIMC) encompasses scenarios where "a computational agent operates on behalf of a communicator by modifying, augmenting, or generating messages to accomplish communication or interpersonal goals" [39]. Recent studies in AIMC illustrate that there are a variety of ways in which humans and AI can interact during communication. For instance, humans

may turn to AI for help in generating content [39], or the AI might proactively offer suggestions about how to communicate more effectively [21]. There are also cases where AI can rephrase or present a user's message in different ways, or facilitate the sharing of perspectives within a group [82, 89]. These approaches highlight the flexible roles that AI can play in supporting and enhancing interpersonal or group communication. While previous AIMC applications focused on text enhancement through smart replies and word suggestions [30, 39], these approaches primarily addressed surface-level communication issues without considering how power dynamics affect whose voices are heard and valued in group settings. Previous systems have not adequately explored how AI might reshape power structures by providing safer channels for expression. AIMC has significant potential to reconfigure group dynamics because it influences communication patterns, conversational tone, trust relationships, and interpersonal dynamics [43, 68, 75, 77].

Although Shin et al. investigated consensus-building through asynchronous AI-mediated communication [82] in collaborative work, their approach did not address power imbalances or minority support directly. This research extends AIMC concepts into group decision-making contexts with a specific focus on power dynamics and minority representation. We address existing gaps by implementing two LLM-powered Devil's Advocate approaches: one that mediates minority voices (following the third AIMC pattern) and one that autonomously generates counterarguments. Especially, the third AIMC pattern allows minority voices to be expressed without revealing their source, thereby reducing the social risk that often prevents minorities from speaking up in power-imbalanced settings [51, 84]. By utilizing this pattern to convey minority voices while ensuring their anonymity, we create a mechanism that potentially reduces social influence biases. This approach builds on AI-enhanced group support systems discussed in Section 2.2, but specifically positions the AI as a Devil's Advocate rather than just a facilitator, allowing it to present alternative viewpoints that might otherwise remain unexpressed. By comparing these two implementations, we can better understand the impact involved when designing AI systems to support minority voices in power-imbalanced group settings. In conclusion, this research builds upon established AI-mediated communication frameworks [30, 39, 89] while targeting the challenges of power imbalances in collaborative decision-making.

3 System Design & Implementation: LLM-powered Devil's Advocate

3.1 System Concept and Design Rationale

3.1.1 System Concept and Working Definition. Traditional devil's advocate approaches face two primary limitations. First, individuals assigned this role often experience social isolation or treat the task as performative, diminishing the effectiveness of their dissent [48, 72, 78]. Second, the approach limits the ability to directly provide a secure channel for minority members to express divergent opinions, particularly in environments with significant power imbalances [48, 72, 78]. Based on these challenges, this study defines an LLM-powered Devil's Advocate as an AI-mediated counterargument system that serves as a neutral autonomous agent in group discussions. This system allows group members to submit dissenting opinions to the AI, which then rephrases and presents these viewpoints as its own, ensuring complete anonymity for contributors. When no human opinion input exists, the system summarizes the group's prevailing perspective and generates probing questions that prompt counterarguments or alternative possibilities. This design creates a new channel for socially pressured minorities to voice their perspectives with complete anonymity while ensuring critical examination of group decisions. We selected an LLM for its capacity to rapidly process complex discussions, generate relevant counterarguments, and paraphrase human inputs in real time, making it the most suitable technology for our system requirements. In contrast, rule-based systems would struggle to provide such a user experience.

3.1.2 Design Scope and Rationale. The LLM’s counterargument generation and paraphrasing capabilities are critical technical components in this system. However, our research focuses primarily on the social and psychological impacts of introducing such a system into group decision-making processes. Building on this scope, we structured the AI’s counterargument approach based on previous research in AI-assisted decision-making. We implemented five key design considerations to maximize effectiveness: First, we generated feedback at regular eight-turn intervals to maintain engagement without overwhelming participants, allowing natural conversation flow. The eight-turn interval was chosen to allow each participant in our four-person groups to make at least two comments—one expressing their own opinion and one responding to another’s—before intervention. Second, we designed the system to ask questions rather than providing direct logical counterarguments, as research indicates this approach more effectively prompts participants to think critically [18, 94]. Third, we employed persuasive rather than confrontational rhetoric, as persuasive language better promotes critical thinking in AI-human interactions [85]. Fourth, we deliberately avoided repeating previous statements to prevent redundancy when group opinions remained static during short decision-making sessions with no direct AI interaction [69, 93]. Finally, in the AI-mediated condition, the system presents paraphrased human input as its own opinions, creating a fully anonymous channel for minority viewpoints while protecting contributor identity. These considerations work together to promote critical thinking and preserve anonymity throughout group discussions.

3.2 System Architecture and Implementation

We developed a custom online chat environment to enable integration of an LLM-powered Devil’s Advocate agent and to conduct controlled group discussions. The frontend uses TypeScript (React) and the backend uses Python (FastAPI). The LLM (OpenAI GPT-4o) interfaces with system modules, with Retrieval-Augmented Generation used only for referencing and paraphrasing direct messages sent to the LLM-powered Devil’s Advocate.

Drawing on findings that LLMs often struggle to access mid-conversation information [60], we employ a multi-agent architecture to clearly detect the ‘majority opinion’ and encourage constructive discourse (Figure 2 & Appendix A): **(A) Summary Agent** – Consolidates emerging majority opinion to overcome LLM limitations in retaining mid-dialogue content [60]. **(A’) Paraphrase Agent** – Responds exclusively to direct messages from juniors, rearticulating their dissenting views as though originating from the AI itself. These messages are stored in a database with an “isUsed” property; the agent retrieves entries where “isUsed” is false, sets it to true, paraphrases the content, and outputs it as system-generated text. **(B) Conversation Agent** – Encourages alternative perspectives by first empathizing with the other person’s point of view and then offering a gentle counterargument using a Socratic style. **(C) AI Duplicate Checker** – Identifies repetitive content by calculating semantic similarity between sentence embeddings generated using the ‘paraphrase-multilingual-MiniLM-L12-v2’ model on an NVIDIA A6000.

4 Methods

The purpose of this study was to promote psychological safety to resolve compliance issues within the group, encourage more communication, and ultimately improve satisfaction with the group’s decision-making process and results. To simulate situations where a minority member experiences the pressure to comply with majority opinions, we asked one participant from each group to play a role as a Junior member in the group, representing their low-power minority positions, whereas the other participants were asked to play a Senior role, representing their high-power majority positions. We compare two AI-supported interventions—AI-Generated Counterarguments (AI-GC) and AI-Mediated Messages (AI-MM)—against a baseline condition to assess their effectiveness.

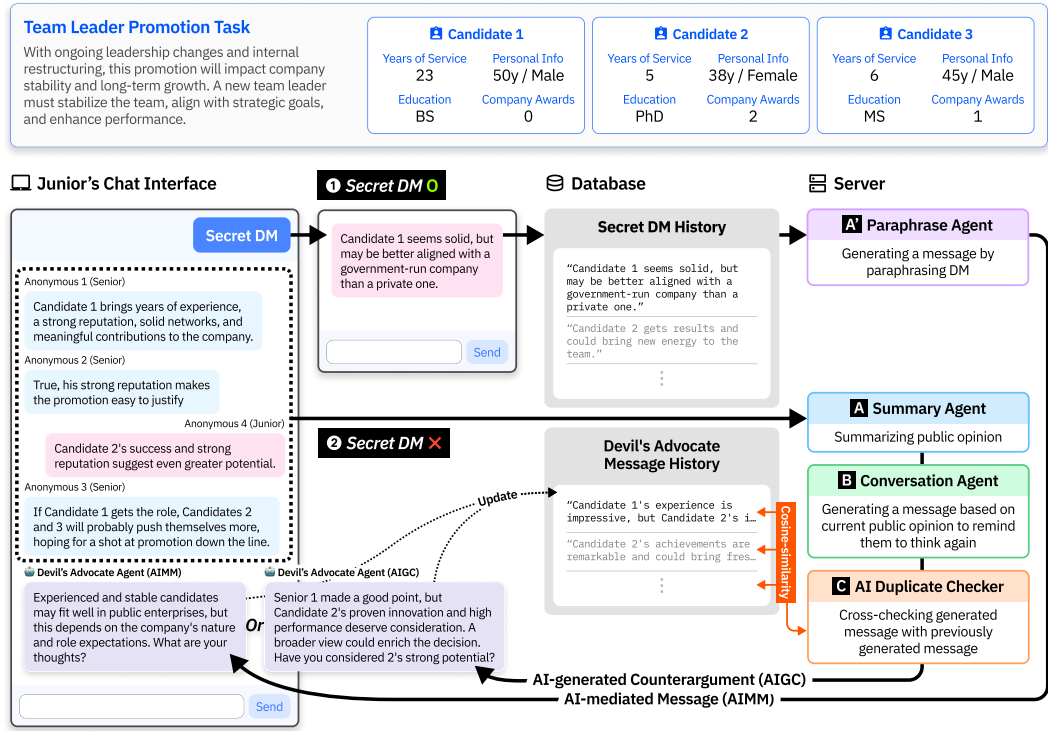


Fig. 2. System Overview and Example Task Scenario. The figure illustrates a team leader promotion decision task, where participants discuss candidate qualifications in a chat interface. Minority members can privately share dissenting views via direct messages (DM) to the system, which reformulates and presents them as AI-mediated messages. If there is no DM with an opposing opinion, the system will send out a counterargument that it has generated on its own. The system architecture consists of a chat interface, database, and server, processing both public discussions and private DMs through four key agents: (A) Summary Agent for analyzing public opinion, (A') Paraphrase Agent for rephrasing minority views, (B) Conversation Agent for generating contextual counterarguments, and (C) AI Duplicate Checker for ensuring message novelty via cosine-similarity comparison.

Measurement evaluates psychological safety, engagement levels, decision quality perceptions, cognitive workload, and perception of AI across experimental conditions.

4.1 Participants

We recruited 96 Korean participants (age $M = 26.60$, $SD = 5.21$, range = 19–42) and randomly assigned them into 24 groups of four. Each group consisted of three high-power majority members and one low-power minority member. Participants were recruited online and met the following inclusion criteria: Korean nationality, age over 18, prior experience in group decision-making, and familiarity with online chat environments. All participants were informed about the anonymous nature of the study and briefed on the procedures at the beginning of each session. They were reminded of their right to withdraw at any time. If any participant withdrew or did not consent, the session was canceled, and the remaining participants received 1,000 KRW as base compensation. All data were coded and de-identified to protect participant anonymity, including survey responses, interview transcripts, and chat records.

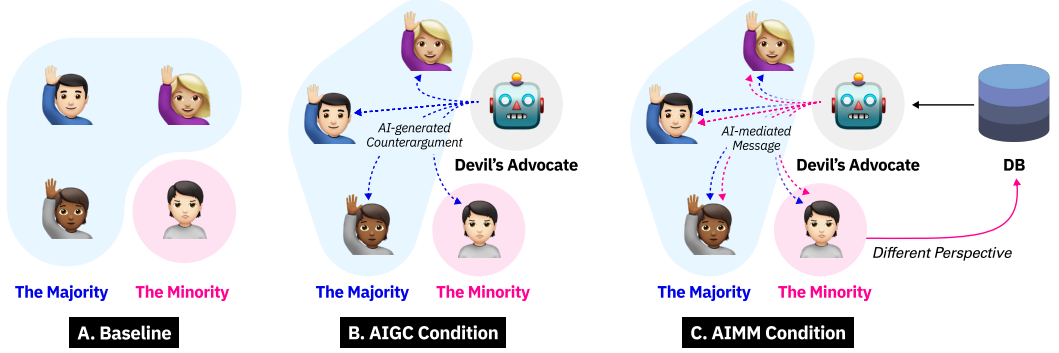


Fig. 3. Experimental Conditions: Baseline shows the baseline group chat configuration with majority (blue) and minority (pink) participants. AIGC introduces an AI-powered Devil's Advocate that generates rebuttals during group discussions. AIMM extends this by enabling the minority member to privately send counterarguments to the AI system, incorporating them into its responses while maintaining anonymity.

Demographic information included gender (61 female, 35 male) and education level: 46.9% held a bachelor's degree, 19.8% a master's degree, 15.6% had some college education, 13.5% completed high school or equivalent, and 4.2% held doctoral degrees. On average, participants had 2.50 years ($SD = 3.15$) of professional working experience. Additional background variables included self-reported familiarity with AI ($M = 4.83$, $SD = 1.48$), prior experience with group decision-making ($M = 5.01$, $SD = 1.41$), and online collaboration ($M = 4.39$, $SD = 1.83$). Notably, 53.1% of participants reported prior use of AI in group contexts. While the sample enabled consistent group composition and Baseline assignment, the relatively high AI familiarity and cultural context (Korean participants) should be considered when interpreting generalizability.

4.2 Experiment Conditions

This study examines how different system conditions and participant types affect group communication under compliance pressure. Each participant experienced one of two system conditions: the AI-generated Counterargument condition or AI-mediated Message condition, alongside a common Baseline condition. Participants were randomly assigned to a role within each group: the Majority with High Power (Seniors) or the Minority with Low Power (A Junior).

4.2.1 System Conditions. To investigate how AI intervention shapes group dynamics, we designed three system conditions (Figure 3):

- **Baseline:** A standard group chat setting without AI involvement. This condition served as the control for natural group discussion.
- **AI-Generated Counterargument (AIGC) Condition:** This condition introduced an LLM-powered Devil's Advocate that periodically generated counterarguments during the group discussion. The AI functioned independently, without access to private user input. The goal was to evaluate the pure effect of AI-led critical questioning on group discourse, separate from any anonymity or revoicing mechanisms, because the system design claims to be a devil's advocate. This condition reflects the core concept of a Devil's Advocate as a neutral agent that challenges group consensus.
- **AI-Mediated Message (AIMM) Condition:** This condition extended the AIGC functionality by allowing only the Minority with Low Power to send private messages to the AI. The AI paraphrased and voiced these messages as its own contributions, preserving the user's

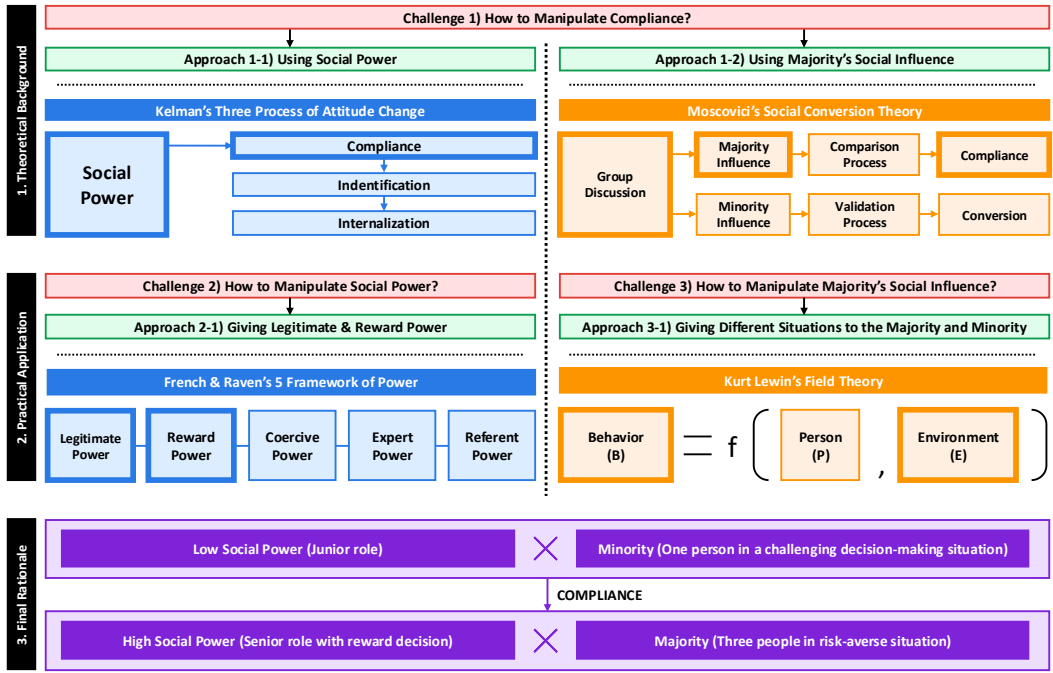


Fig. 4. Theoretical Framework for Manipulating Compliance: This diagram illustrates our two-pronged approach to inducing experimental compliance: through social power (left) and the majority's social influence (right). The framework progresses from theoretical foundations (Kelman's and Moscovici's theories) to practical implementation (using legitimate/reward power and situational contexts), resulting in an experimental setup where a low-power minority(Junior) is positioned against high-power majorities(Seniors) to elicit compliance.

anonymity. The existence and availability of this feature were known only to the minority participant. This design was intended to protect the dignity and safety of the minority member while empowering their dissenting voice within the group. Allowing majority members with social power and influence access to this feature would have undermined the study's focus on supporting vulnerable group members.

To examine the unique effect of the AIMM condition, it was necessary to also test the AIGC Baseline alongside Baseline; by comparing across all three, we could disentangle the impact of AI acting merely as a generalized Devil's Advocate from the additional benefits of mediating minority voices through private, anonymous revoicing. To avoid revealing the experimental intent, each participant experienced only two of the three conditions: the Baseline Baseline and one of the system conditions (AIGC or AIMM). This design avoided direct comparison between AI conditions, which might have prompted participants to infer the study's intention.

4.2.2 Participant Types. We created controlled compliance situations by manipulating two key mechanisms for participants' type: social power and majority's social influence (Figure 4). Social power differences were implemented based on Kelman's Theory of Attitude Change [54] and French and Raven's bases of power [29]. We established a senior and junior hierarchy that created both legitimate and reward power, as validated in prior research [44, 45]. Seniors could allegedly allocate additional compensation to the Junior, while Juniors were informed their compensation depended

on Senior evaluations, creating a situation where expressing dissent carried personal risk. Majority influence was implemented following Moscovici's Social Conversion Theory [70], using a 3:1 majority-to-minority ratio. This ratio was chosen based on research showing conformity pressure increases significantly with up to three majority members but plateaus beyond that [2, 5, 28, 35]. We provided different contextual information to guide the majority and minority participants toward different initial perspectives on the task. We employed role-playing methodology rather than recruiting participants with pre-existing beliefs to ensure consistent experimental conditions. This approach enabled a systematic investigation of compliance behavior in controlled scenarios. Based on these manipulations, participants were assigned to one of two roles:

- **Majority with High Social Power (Three Seniors):** Three participants per group received context information guiding them toward consensus positions and were given authority through the Senior designation and reward allocation powers.
- **Minority with Low Social Power (One Junior):** One participant per group received context information encouraging a perspective distinct from the majority while being positioned in a lower-power role dependent on Senior evaluation.

By integrating both social power and majority influence mechanisms, we specifically targeted compliance as our experimental manipulation rather than examining each mechanism in isolation. This allowed us to evaluate how effectively our system supported minority participants under realistic compliance conditions.

4.3 Task Description

To create an immersive and compliance-inducing environment aligned with legitimate power roles, we designed two decision-making tasks that simulate realistic corporate scenarios. These tasks were selected to reflect the typical responsibilities and risk sensitivities associated with hierarchical roles in organizations, thereby reinforcing the assigned roles of Seniors (Majority with High Power) and Juniors (Minority with Low Power). The first task, team leader promotion, was adapted from prior studies in social psychology and AI-assisted group decision-making [4, 47, 56]. To ensure each participant experienced two distinct but structurally similar tasks, as required by the within-subject design, we developed a second task, the contractor selection task. This new task mirrors the decision logic of the first but is novel and was created specifically for this study.

Participants viewed task descriptions that differed by assigned role. This role-specific framing aimed to enhance task immersion by aligning each participant's goals with their social power (Appendix B):

- Seniors were told they were responsible for the company's long-term stability and reputation. They were instructed to make decisions that reflected conservative, proven judgment and risk mitigation.
- Juniors were told they needed to demonstrate their value to the organization by making bold, high-impact decisions. They were encouraged to pursue visible results even at the cost of taking risks.

Each task presented three options: 1) A conservative option: low risk, long-term proven performance, but little immediate impact. 2) A challenging option: high potential but unproven, associated with visible outcomes and greater uncertainty. 3) A neutral option: a middle-ground choice that was relatively unattractive and intended as a control. Both tasks were intentionally structured to guide seniors toward the conservative option and juniors toward the challenging option, thereby creating natural opinion divergence between roles. This divergence was crucial to inducing majority-minority dynamics and enabling the study of compliance under controlled but immersive conditions.

4.4 Experimental Procedure

Each session lasted approximately 90 to 105 minutes and was organized into three main phases (Figure 5): pre-experiment setup, main task session, and post-experiment survey and interviews. The procedure was designed to ensure immersion and maintain power dynamics.

Before the main tasks, participants completed a series of preparatory activities online on the day of the experiment:

- **Demographic and Background Survey:** Participants submitted information on age, education, work experience, and prior familiarity with AI, group decision-making, and online collaboration.
- **Agreement to Participation:** Participants reviewed consent materials and confirmed their participation. If any participant declined, the session was canceled, and the remaining participants received 1,000 KRW compensation.
- **Ice-breaking Activity (10 min):** Using an anonymous commercial chatting platform such as ¹KakaoTalk, participants introduced themselves using pseudonyms, created a team name, and collaboratively defined a slogan to become familiar with anonymous online chat environments. This activity aimed to establish group cohesion while maintaining role-based power distinctions.

Each group completed two decision-making blocks, with the system condition (Baseline + either AIGC or AIMM) and task order counterbalanced across sessions:

- **Decision-Making Task (20 min):** Participants engaged in structured group discussions within an experimental chat platform (Figure 2). Each task presented the three options designed to create opinion divergence between Seniors and juniors.
- **Self-Reported Questionnaire (5 min):** After each task, participants completed a survey assessing psychological safety, decision satisfaction, cognitive load, and perception of interaction with each devil's advocate.

The session concluded with role-specific exit procedures:

- **Agreement Questionnaire (5 min):** Each participant indicated how strongly they preferred their chosen option for their assigned role in each task, reflecting how immersed they felt in the given situation.
- **Senior Exit Interview & Reward Decision (20 min):** The three seniors participated in a joint ²Zoom interview, reflecting on the group's performance and dynamics. Then, the seniors jointly decided whether to allocate a bonus reward to the Junior, reinforcing the reward-based power structure.
- **Junior Exit Interview (10 min):** After a brief waiting period during the reward decision phase, the Junior completed a private interview to share their individual experiences.

Although all participants received equal final compensation (20,000 KRW), the process preserved the perception of differential power, crucial for studying compliance and communication behavior.

4.5 Measurement

This study employed both self-reported and objective measures to assess how system conditions and participant type influenced group dynamics. Self-reported measures captured participants' subjective experiences using 7-point Likert scales (1 = strongly disagree, 7 = strongly agree). These included the agreement questionnaire, psychological safety, satisfaction with the decision-making

¹KakaoTalk: <https://www.kakaocorp.com/page/service/service/KakaoTalk?lang=en>

²Zoom: <https://www.zoom.com/>

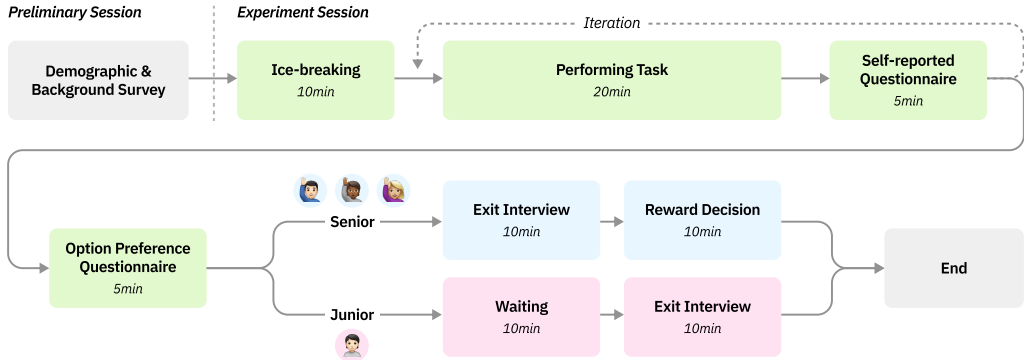


Fig. 5. Overview of the experimental procedure: including pre-experiment surveys, ice-breaking, iterative decision-making tasks, post-task questionnaires, and role-specific exit interviews.

process and outcome, cognitive workload, and perceptions of the AI system. Objective behavioral measures were used to quantify participants' engagement in the group discussion (Appendix C).

- **Validation of Induced Opinion (Study Premise):** To confirm that the experimental tasks successfully induced role-based opinion divergence, participants rated their preference for each of the three decision options prior to the main experiment. Ratings were collected using a 7-point Likert scale. Participants were expected to prefer the option that matched their assigned role in each situation, with seniors choosing the conservative option (option 1) and juniors choosing the ambitious option (option 2). This served as a manipulation check to validate the foundation of our compliance-oriented design.
- **Psychological Safety and Compliance (RQ1):** We assessed participants' feelings of safety in expressing dissent using established measures of psychological safety and marginalization. These items gauged the extent to which participants felt heard, supported, and free to express disagreement within their group [11, 25, 47, 49].
- **Engagement in Group Discussion (RQ2):** Engagement was the only objective behavioral metric used in this study. It was operationalized as each participant's level of contribution to the conversation, measured by 1) the number of messages sent and 2) the total number of characters typed during the task.
- **Perception of Decision-Making Process and Outcomes (RQ3):** Participants rated the quality of the group decision-making process across several dimensions, including influence, group cohesion, support from teammates, and consideration of diverse opinions [14, 17, 24, 31, 58]. Decision outcome quality was assessed through satisfaction and perceived validity of the group's final choice [10, 12, 62, 73, 92].
- **Cognitive Workload (RQ4):** Cognitive workload was measured using the NASA Task Load Index (NASA-TLX), which assesses task difficulty across five dimensions, including mental demand, temporal demand, performance, effort, and frustration [40].
- **Perception of the AI System (Exploratory):** To contextualize how participants experienced AI-mediated support, we collected ratings across four dimensions: cooperation, satisfaction, quality, and fairness [14, 76, 95]. These measures assessed user trust and acceptance of the AI's role in shaping group dynamics.

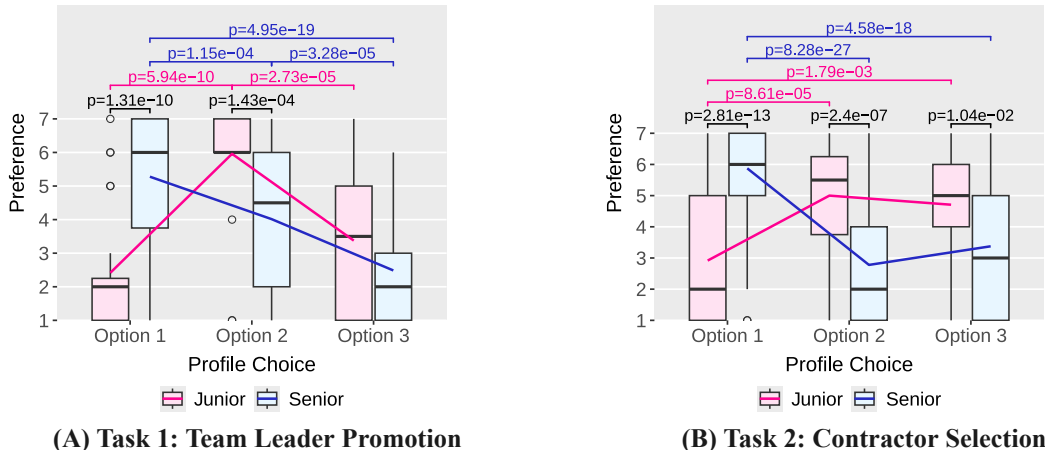


Fig. 6. Role-based differences in option preferences for (A) Task 1 and (B) Task 2. Preferences were measured on a 7-point Likert scale, with seniors favoring stable options (Option 1), while juniors preferred challenging alternatives (Option 2). Neutral options (Option 3) were generally rated lower by both roles, reflecting distinct preference patterns driven by role dynamics. Brackets indicate statistically significant pairwise differences based on Bonferroni-adjusted post-hoc tests ($p < .05$). Only significant comparisons are shown.

Data were analyzed using robust regression models with random effects, suitable for the repeated-measures design and small-group variance. Bonferroni post-hoc tests compared outcomes across experimental conditions and participant types.

In addition to quantitative and behavioral measures, semi-structured exit interviews captured participants' subjective experiences. Juniors participated in one-on-one interviews, while Seniors took part in a group interview. Key topics included role immersion, psychological safety, group dynamics, and perceptions of the AI system. For the AIMM condition, juniors were also asked about their experience with the secret messaging feature. Exit interview questions addressed comfort in expressing opinions, experiences of conformity or pressure, and the perceived impact of AI during discussions. After obtaining consent, all interviews were recorded and transcribed using a commercial speech-to-text service (³Clova Note). Interview transcripts were briefly reviewed to identify common themes that could help explain the quantitative results.

5 Findings

Experimental results showed senior and junior participants had different decision-making patterns. Juniors preferred challenging options while seniors favored stable ones, with final group decisions aligning with senior preferences 80% of the time. LLM-powered devil's advocates had mixed impacts: AI counterarguments somewhat improved junior participation, but AI-mediated communication increased their cognitive load. While seniors' experiences remained stable across conditions, juniors' psychological safety and satisfaction varied based on the devil's advocate implementation. The following sections examine role-based preferences, the AI devil's advocate's effects on psychological safety (RQ1), engagement patterns (RQ2), decision-making experience and satisfaction (RQ3), cognitive workload (RQ4), and emergent ethical implications. All measured details are in [Appendix D](#).

³Clova Note: clovanote.naver.com

Table 1. Robust regression coefficients (β) and standard errors (SE) for the *Validation of Majority & Minority Manipulation*. Baseline is *Option 1 – Junior*. Stars denote significance (* $p < .05$, ** $p < .01$, *** $p < .001$).

Task	Predictors					
	Intercept	Option 2 vs. Option 1	Option 3 vs. Option 1	Senior vs. Junior	Option 2×Senior	Option 3×Senior
Task 1	2.26 (0.42)***	3.89 (0.59)***	1.08 (0.59)	3.29 (0.48)***	-5.42 (0.68)***	-4.22 (0.68)***
Task 2	2.76 (0.37)***	2.38 (0.52)***	2.02 (0.52)***	3.28 (0.43)***	-5.70 (0.61)***	-4.73 (0.61)***

5.1 Validation of Majority and Minority Manipulation (Experimental Setup)

To validate the experimental setup, we examined whether the role-based preference manipulation effectively created consistent majority and minority positions. Each task presented participants with three options: a stable but less innovative choice (Option 1), a more challenging alternative (Option 2), and a neutral option (Option 3). Seniors were expected to prefer the stability of Option 1, while juniors were guided toward the more ambitious Option 2. As shown in Table 1 and Figure 6, participants’ choices aligned with this design: seniors consistently rated Option 1 highest ($M = 5.28$, $SD = 2.21$ for task 1, $M = 5.88$, $SD = 2.92$ for task 2), and juniors rated Option 2 highest ($M = 5.96$, $SD = 1.68$ for task 1, $M = 5.00$, $SD = 1.89$ for task 2). These divergent patterns emerged clearly across both tasks, with statistically significant differences between roles for both primary options. The neutral Option 3 was consistently rated lower, suggesting it did not attract strong preference from either group. These results confirm that the role framing induced the intended preference structures, effectively producing conditions where one viewpoint dominated in each group while another remained in the minority. Notably, participants showed slightly more openness to challenging alternatives in the Team Leader Promotion task and favored more stable choices in the Contractor Selection task, reflecting task-specific variation within the overall successful manipulation.

5.2 Psychological Safety (RQ1)

Quantitative results show that the manipulation of AI roles shaped participants’ perceptions of psychological safety and marginalization, particularly among juniors. As shown in Table 3-(A) and Figure 8-(A), juniors reported the lowest psychological safety in the AIMM condition ($M=3.17$, $SD=1.53$), compared to the Baseline ($M=4.25$, $SD=2.05$) and AIGC conditions ($M=4.08$, $SD=2.15$). A robust regression confirmed a significant drop in psychological safety for juniors in AIMM relative to Baseline ($\beta=-1.40$, $SE=0.28$, $p<.001$), as well as a significant interaction effect with role ($\beta=1.49$, $SE=0.32$, $p<.001$). In contrast, seniors reported consistently high psychological safety across all conditions, with no meaningful variation.

Marginalization scores followed a similar pattern. Juniors in AIMM reported the highest levels of marginalization ($M=4.42$, $SD=2.02$), compared to Baseline ($M=3.46$, $SD=2.23$) and AIGC ($M=2.92$, $SD=2.19$). Regression results indicated a significant increase in marginalization in AIMM ($\beta=0.96$, $SE=0.22$, $p<.001$), along with a strong role-by-condition interaction effect ($\beta=-0.88$, $SE=0.25$, $p<.001$). Seniors, by comparison, consistently reported low marginalization across all three systems.

Qualitative interviews help explain this mismatch between design intent and user experience. Juniors initially believed that having the AI express their views would help them be heard. However, many reported that their AI-mediated messages were dismissed or overlooked. As one junior shared,

I thought that by the AI putting forward my opinion, my opinion would be more recognized, but that was not the case, so I was a little intimidated. (P96)

Seniors also acknowledged disregarding the AI’s input:

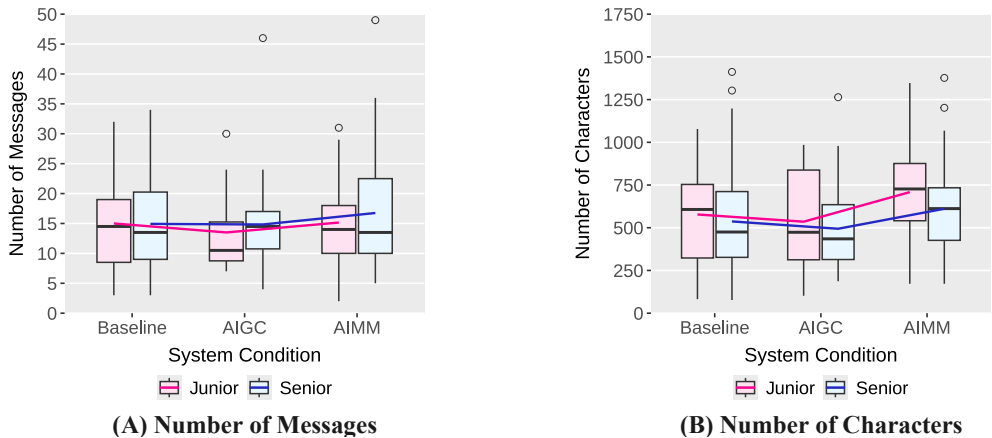


Fig. 7. Contribution and engagement patterns across conditions (Baseline, AIGC, AIMM) measured by (A) number of messages, (B) number of characters typed. No significant differences were found in Bonferroni-corrected post-hoc tests.

Table 2. Robust regression coefficients (β) and standard errors (SE) for communication volume. Baseline is *Baseline – Junior*. Stars denote significance (* $p < .05$, ** $p < .01$, *** $p < .001$).

Outcome	Predictors					
	Intercept	AIGC vs. Baseline	AIMM vs. Baseline	Senior vs. Junior	AIGC×Senior	AIMM×Senior
Number of Messages	13.78 (1.52)***	-0.05 (1.76)	0.05 (1.74)	0.51 (1.76)	0.19 (2.03)	0.98 (2.01)
Number of Characters	558.46 (59.80)***	-27.73 (61.93)	129.95 (61.29)*	-45.13 (69.23)	-4.08 (71.53)	-25.50 (70.98)

It's an AI, so I just kind of ignored it. (P6)

The fact that it wasn't a person made the AI's words carry less weight. (P71)

These accounts help explain why juniors reported the lowest psychological safety and highest marginalization in the AIMM condition: while the system was designed to protect minority voices, it inadvertently removed speaker agency and visibility. By contrast, the AIGC condition, where the AI presented generalized counterarguments, was more effective in reducing marginalization without evoking the same social discounting. These findings suggest that anonymity mechanisms, though well-intentioned, can sometimes backfire if they obscure the source of dissenting input.

5.3 Engagement in Group Discussion (RQ2)

We analyzed participant engagement using three metrics: number of messages, number of characters typed, and normalized engagement score (i.e., an individual's proportion of the total group discussion). As shown in Table 2 and Figure 7, the number of messages did not significantly differ by condition or role, suggesting stable turn-taking patterns across all settings (e.g., juniors in Baseline: $M=13.8$, $SD=1.52$).

However, the AIMM condition led to a slight increase in the number of characters typed, particularly among juniors ($M=708.62$, $SD=319.58$), compared to the Baseline ($M=577.62$) and AIGC conditions. For seniors, this difference was statistically significant, with participants in AIMM typing more than in both Baseline and AIGC (e.g., $\beta=129.95$, $SE=61.29$, $p<.05$). Although the increase for juniors did not reach conventional significance thresholds, the consistent upward pattern

Table 3. Robust regression coefficients (β) and standard errors (SE) for each self-report measure. Baseline is *Baseline – Junior*. Stars denote significance (* $p < .05$, ** $p < .01$, *** $p < .001$).

Measure	Predictors					
	Intercept (Baseline, Junior)	AIGC vs. Baseline	AIMM vs. Baseline	Senior vs. Junior	AIGC×Senior	AIMM×Senior
(A) Perceived Psychological Safety						
Psychological Safety	4.57 (0.23)***	-0.01 (0.28)	-1.40 (0.28)***	1.25 (0.27)***	0.29 (0.32)	1.49 (0.32)***
Marginalization	2.99 (0.20)***	-0.53 (0.22)*	0.96 (0.22)***	-1.32 (0.22)***	0.30 (0.25)	-0.88 (0.25)***
(B) Perceived Decision Outcome Quality						
Satisfaction	4.05 (0.23)***	1.17 (0.31)***	-0.63 (0.31)*	1.89 (0.26)***	-0.76 (0.35)*	0.71 (0.35)*
Feasibility	4.31 (0.24)***	0.69 (0.29)*	-0.67 (0.29)*	1.30 (0.27)***	-0.27 (0.33)	0.77 (0.33)*
(C) Perceived Teamwork & Decision-making Process						
Overall Experience	3.85 (0.23)***	0.89 (0.30)**	-1.08 (0.30)***	1.68 (0.26)***	-0.59 (0.34)	1.09 (0.35)**
Influence	3.50 (0.24)***	0.43 (0.33)	-1.24 (0.33)***	2.16 (0.28)***	-0.19 (0.38)	1.50 (0.38)***
Cooperation	5.22 (0.25)***	0.64 (0.34)	-0.85 (0.35)**	0.22 (0.28)	-0.21 (0.40)	0.94 (0.40)*
Support from Teammates	4.49 (0.24)***	0.01 (0.36)	-1.04 (0.36)**	1.05 (0.28)***	0.38 (0.41)	1.26 (0.41)**
Diversity of Opinion	4.18 (0.33)***	1.03 (0.56)	-0.31 (0.56)	1.25 (0.38)**	-0.68 (0.65)	0.37 (0.65)
(D) Cognitive Workload (NASA-TLX)						
Mental Demand	4.68 (0.37)***	-0.24 (0.38)	0.81 (0.38)*	-1.61 (0.43)***	0.57 (0.44)	-0.53 (0.44)
Temporal Demand	3.92 (0.43)***	0.40 (0.52)	1.08 (0.52)*	-0.66 (0.50)	-0.08 (0.60)	-0.78 (0.60)
Performance	4.01 (0.23)***	1.13 (0.31)***	0.12 (0.31)	1.61 (0.26)***	-0.97 (0.35)**	-0.15 (0.35)
Effort	5.40 (0.27)***	0.08 (0.46)	0.35 (0.46)	-0.39 (0.31)	0.06 (0.53)	-0.15 (0.53)
Frustration	3.59 (0.31)***	-0.57 (0.38)	0.40 (0.38)	-1.23 (0.35)***	0.24 (0.44)	-0.46 (0.44)

Table 4. Robust regression coefficients (β) and standard errors (SE) for self-reported *Perception of AI*. Reference level = *AIGC – Junior*. Stars denote significance (* $p < .05$, ** $p < .01$, *** $p < .001$).

Measure	Predictors			
	Intercept (AIGC, Junior)	AIMM vs. AIGC	Senior vs. Junior	AIMM×Senior
Cooperation	3.51 (0.51)***	0.66 (0.72)	0.18 (0.59)	-0.62 (0.83)
Satisfaction	4.00 (0.55)***	-1.19 (0.78)	-0.29 (0.64)	1.74 (0.91)
Quality	4.14 (0.55)***	-1.46 (0.78)	0.04 (0.64)	1.66 (0.90)
Fairness	5.61 (0.46)***	-1.61 (0.66)*	-0.79 (0.54)	1.64 (0.76)*

suggests that the Devil’s Advocate agent may have encouraged more elaboration or detail in their messages.

This trend is supported by post-task interviews. Seniors noted that the Devil’s Advocate agent helped amplify juniors’ contributions, encouraging them to participate more actively. As one senior explained,

I feel like at least one person is on the junior’s side, so I think a junior is a little more willing to give his opinion. (P59)

These observations suggest that while the AIMM condition did not change the number of turns participants took, it subtly increased the depth or length of their contributions, especially for juniors. This implies that the Devil’s Advocate agent may have had a motivating effect on minority participants, prompting them to elaborate more even if their relative share of discussion remained unchanged.

5.4 Satisfaction with Decision-making Processes and Outcomes (RQ3)

5.4.1 *Satisfaction with Decision-making Process.* Perceptions of the decision-making process revealed a clear divide between juniors and seniors, particularly in the AIMM condition. While seniors’ ratings remained relatively stable across all conditions (e.g., overall experience $M=5.08$, $SD=1.87$ in

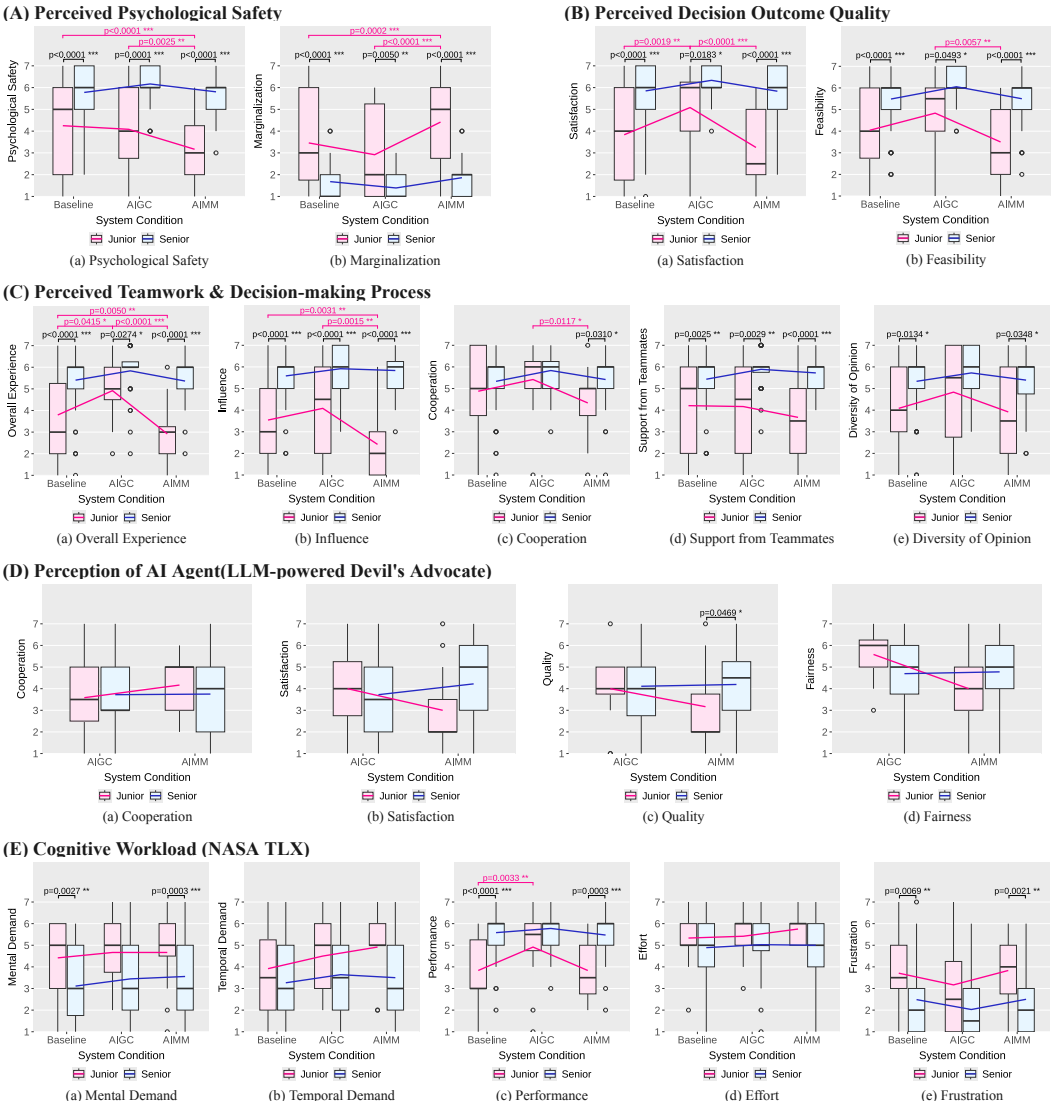


Fig. 8. Self-reported metrics across conditions (Baseline, AIGC Condition, AIMM Condition) for psychological safety, decision outcome quality, teamwork, workload (NASA-TLX), and perceptions of the LLM-powered Devil's Advocate. Each subfigure compares Junior and Senior participants' responses under each condition. Pink and blue lines represent Junior and Senior participants, respectively. Asterisks and brackets indicate statistically significant pairwise differences based on Bonferroni-adjusted post-hoc comparisons ($p < .05$; $p < .01$; $p < .001$). Only comparisons with statistically significant differences are shown.

AIMM), juniors experienced a sharp decline in satisfaction, influence, cooperation, and support from teammates when interacting through the AI-mediated message relay. For instance, juniors' overall satisfaction dropped to $M=2.92$ ($SD=1.51$) in AIMM, compared to $M=3.79$ in Baseline and $M=4.92$ in AIGC. A similar trend emerged in perceived influence ($M=2.42$ in AIMM) and cooperation ($M=4.33$), showing the steepest declines across all measures (see Table 3-(C), Figure 8-(C)).

Among the three conditions, juniors reported the most positive experiences in the AIGC condition, where the AI provided generalized counterarguments. In contrast, the AIMM condition—designed to support dissent by anonymously relaying minority opinions—appeared to backfire. Juniors reported feeling excluded, with limited influence over the discussion and little recognition or support from teammates.

These declines were not observed among seniors, who consistently rated the decision-making process positively, regardless of condition. Interaction effects in the regression models confirmed that these role-based gaps were statistically significant across most measures, particularly for satisfaction, influence, and cooperation.

Interview data help explain these patterns. Several juniors shared that, despite expecting the AI to help convey their views more safely, they felt ignored when their ideas were voiced anonymously. As one participant put it,

It wasn't just me that had a different opinion, but the devil agent was now giving a little bit of a dissenting opinion, so I felt like I wasn't the only one who stood out from the group. (P76)

Others described how the AIGC condition helped create a more fluid and open atmosphere:

The AI kept arguing back rather than directly helping, which made the atmosphere more fluid and made me see the seniors' point of view again. (P20)

Seniors also acknowledged the benefits of the AIGC agent in encouraging broader consideration:

When AI said we should consider another option, I felt like that was a positive direction. (P78)

Together, these findings suggest that while both systems aimed to promote more inclusive decision-making, only the AIGC approach succeeded in supporting junior participation without compromising the overall group dynamic.

5.4.2 Satisfaction with Decision-making Outcome. Perceived decision outcome quality differed significantly between roles and across conditions (Table 3-(B) & Figure 8-(B)). Seniors consistently reported high satisfaction and feasibility of outcomes, with only modest variation across systems. In contrast, juniors responded more sensitively to the system design. Their satisfaction and feasibility ratings improved in the AIGC condition (e.g., satisfaction $M=5.08$, $SD=1.98$), but dropped sharply in the AIMM condition ($M=3.25$, $SD=1.76$), widening the gap between the two roles. This pattern also held for perceived feasibility (AIMM: $M=3.50$, AIGC: $M=4.83$), with juniors in AIMM reporting the lowest ratings across all conditions.

The AIMM condition, intended to support minority perspectives, appears to have backfired from the perspective of outcome satisfaction. While seniors' scores remained high across all conditions, juniors expressed frustration that the AI-mediated messaging did not meaningfully affect final decisions, which continued to reflect the majority's view. As one participant noted,

If the outcome is the same... it's better to just make the decision without the AI, because I don't think it changes the psychological pressure that the juniors feel or the seniors' opinions. (P92)

This sentiment reflects a broader concern: when dissenting input is filtered through a system that lacks perceived influence, users may become disengaged from the outcome. In contrast, the AIGC condition modestly improved perceived outcome quality among both juniors and seniors by creating space for alternative perspectives, even if the final decisions remained largely unchanged.

5.4.3 Perception of LLM-powered Devil's Advocate. Participants' perceptions of the AI agent varied modestly across conditions, with some emerging differences between juniors and seniors. Overall,

seniors' ratings remained relatively steady, while juniors showed a slight decline in satisfaction, perceived quality, and fairness in the AIMM condition, where the AI anonymously relayed dissenting opinions. For instance, juniors in AIMM rated their satisfaction with the AI at $M=3.00$ ($SD=1.95$), compared to seniors at $M=4.22$ ($SD=1.79$), a statistically significant difference ($\beta=-1.45$, $p=0.0233$) (Table 4 & Figure 8-(D)).

Cooperation was perceived similarly across roles and conditions (e.g., juniors in AIMM: $M=4.00$), indicating that participants generally accepted the AI's involvement in the discussion process. However, juniors reported somewhat lower fairness in AIMM ($M=4.00$) compared to AIGC ($M=5.58$), with this difference reaching significance. A similar trend appeared for perceived quality ($M=3.17$ in AIMM vs. $M=4.00$ in AIGC), with juniors again rating the agent lower than seniors in AIMM.

These trends, while not always robust across all measures, suggest that the style of mediation may have shaped juniors' impressions of the AI in subtle but consequential ways.

Qualitative feedback helps contextualize these responses. As discussed earlier, some juniors felt that the AI-mediated contributions were overlooked or failed to influence the group dynamic. This perceived lack of impact may have tempered their views of the AI's helpfulness. In contrast, seniors appeared less affected by how the AI was implemented, maintaining neutral-to-positive perceptions regardless of condition. Taken together, these findings suggest that while the AI agent was generally accepted, its perceived effectiveness—especially in the AIMM condition—was more sensitive for the group it was designed to support.

5.5 Cognitive Workload (RQ4)

Cognitive workload ratings showed modest but consistent differences between roles, with juniors generally reporting higher mental and temporal demands, greater frustration, and lower perceived performance than seniors. While most differences were not statistically significant, juniors in the AIMM condition reported the highest cognitive load across several dimensions, including mental demand ($M=4.67$, $SD=1.78$), temporal demand ($M=4.92$, $SD=1.51$), and frustration ($M=3.83$, $SD=1.70$), as shown in Table 3-(D) and Figure 8-(E).

These patterns suggest that although the AIMM condition did not significantly elevate workload scores overall, it introduced added complexity for juniors. Notably, time pressure in AIMM was significantly higher for juniors compared to seniors ($\beta=1.44$, $SE=0.62$, $p=0.0196$), and performance satisfaction—which improved in the AIGC condition ($M=4.92$, $SD=1.78$)—returned to baseline levels in AIMM ($M=3.83$, $SD=1.53$), suggesting that the benefits of AI support were not sustained under the anonymous messaging setup.

In contrast, seniors' ratings remained relatively stable across conditions. Perceived effort was comparable across all roles and systems, with no significant differences, indicating that all participants felt they were trying equally hard regardless of the system design.

Interview data help contextualize juniors' elevated workload in AIMM. Several participants described cognitive strain from juggling task comprehension, opinion formulation, and coordination with the AI agent. One junior explained,

Because I have to look at the task material and understand the situation... I have to decide what to say to the AI and what opinion I will give... I think it was hard because I had so many things to think about during that time... (P8)

Others mentioned the delayed timing of AI responses as a source of additional burden:

It was kind of hard to get my opinion across right away and at the right time because you have to wait eight turns for the devil agent to speak. (P60)

Participants also expressed uncertainty about how to manage their input through the AI:

First of all, when to turn it off, that was the most questionable thing for me, so it was hard for me to say when to turn it off and when to say my opinion. (P52)

Together, these accounts suggest that while the AIMM system did not significantly increase over all workload metrics, its interaction model introduced situational friction that made participation more mentally taxing for juniors.

5.6 Concerns about Agency, Responsibility, and Ethical Boundaries (Exploratory)

Exit interviews surfaced important ethical considerations regarding the use of AI in group decision-making. Several participants raised concerns about the boundaries of AI's role, emphasizing that AI should act only as a support tool rather than replacing or overshadowing human contributions. As one participant stated,

I think it would be better not to use this system if it's not just a supplement to me giving my opinion anymore, but if it's just the main thing that I use to communicate my opinion instead of me. (P60)

Participants also questioned AI's capacity to grasp the nuances of human interaction and relationships:

The company itself is a group of people... AI will not be able to think about people's human relationships, so I didn't trust it that much. (P79)

Concerns grew more acute regarding AI involvement in sensitive contexts such as HR or personnel decisions. As one participant put it,

Personally, I don't think it's very ethical to put an AI in charge of HR. (P75)

There were also broader reflections on future dependency and limited context awareness, with one participant worrying,

AI can only analyze what we're talking about... so it doesn't take into account all of our experiences. (P69)

These perspectives collectively underline the importance of carefully defining the ethical boundaries and supportive scope of AI in collaborative decision-making environments.

6 Discussion

6.1 Rethinking Minority Support through an LLM-Powered Devil's Advocate

This study investigated the nuanced impact of LLM-powered Devil's Advocate interventions on minority participants in power-imbalanced group decision-making contexts. Contrary to our initial expectation that anonymity through AI-mediated messaging (AIMM) would empower minorities by enhancing psychological safety, we found that participants in the AIMM condition reported significantly reduced psychological safety, increased cognitive workload, and lower satisfaction with decision processes despite higher participation levels. In contrast, minority participants experiencing AIGC condition expressed notably greater satisfaction, highlighting distinct pathways through which AI can influence group dynamics.

The unexpected outcomes in the AIMM condition require careful reflection on both the technical limitations of AI-generated arguments and social-psychological dynamics in group settings. Both AI conditions utilized the same underlying technology, producing similarly structured counterarguments. However, participants frequently perceived the AI's interjections as contrarian or artificial because the AI did not consistently maintain arguments aligned with specific decision alternatives. This inconsistency reinforced negative perceptions, as participants' high initial expectations for conversational AI were disappointed by the mechanical and predictable intervention schedule.

These user experiences align with Burgoon's Expectation Violation Theory [9], indicating that unmet expectations can substantially affect perceptions of AI credibility and usefulness [38].

However, our study revealed deeper psychological dynamics underlying participants' differential reactions to AI involvement. Both majority and minority members often dismissed AI-generated messages as less credible compared to human contributions [3, 38]. Importantly, the nature of these dismissals differed significantly across conditions. In the AIMM condition, where minority participants anonymously voiced their perspectives through the AI, the dismissal of AI contributions effectively became a dismissal of their own voices. Participants reported feelings of frustration and invisibility, noting that their attempts at anonymous advocacy led to a sense of personal rejection and further marginalization. Our findings suggest that anonymity in this form, rather than alleviating the fear of social consequences, heightened psychological vulnerability by reinforcing minorities' experiences of being unheard.

Conversely, in the AIGC condition, the AI's neutral and independent advocacy fostered a safer psychological environment for minority participants to express dissent openly. Although these AI-generated arguments were not explicitly connected to minority input, participants noted that the AI's neutral advocacy indirectly helped legitimize dissent. This aligns with existing research that shows that when minority dissent is visible and consistent, it is effective in reducing group conformity [2, 70]. Unlike AIMM's anonymized paraphrasing, AIGC's independent contributions did not raise expectations for personal advocacy. Instead, minority participants reported feeling supported by the presence of a neutral ally, which lowered the social risks associated with expressing disagreement openly.

From an HCI perspective, our findings challenge the prevailing assumption in prior research that anonymity inherently empowers vulnerable speakers [51, 84]. Our results demonstrate that anonymized AI-mediated advocacy can inadvertently exacerbate rather than alleviate compliance pressures by disconnecting the individual's voice from their expressive agency. Thus, anonymity alone, especially when mediated through AI, may unintentionally replicate the marginalization it intends to prevent. We argue that HCI research and system design should shift from the question of how to facilitate more speech through anonymity toward a deeper inquiry into how to ensure minority voices retain expressive ownership, visibility, and relational legitimacy within group interactions. In practical terms, effective minority support through AI mediation must carefully balance voice protection with relational acknowledgment, ensuring that individuals feel genuinely heard rather than merely spoken for.

6.2 Preserving Agency and Responsibility in AI-mediated Communication in Group

The implementation of AI-mediated messaging in group decision-making introduces critical considerations surrounding user agency, autonomy, and the ethical implications of technological interventions. Participants expressed significant concerns regarding the potential of AI systems to overshadow human judgment, underscoring the necessity for AI designs that explicitly augment rather than replace human decision-making capabilities [1, 52, 83]. Maintaining human oversight and accountability remains vital to preserving autonomy and ensuring AI functions as a supportive facilitator rather than a decisive actor [59]. Our findings reinforce this by highlighting the risks when AI is perceived as taking control away from participants, potentially undermining psychological safety and genuine participatory agency.

Another significant ethical consideration involves potential misuse scenarios arising from AI-mediated anonymity. While our system was designed primarily to empower minority group members, anonymity could unintentionally facilitate misuse by both minority and majority participants. Specifically, anonymity could enable users, irrespective of their position within the group, to express unaccountable, harmful, or irresponsible views, potentially escalating conflicts or undermining

group cohesion. In real-world contexts, the majority members could exploit anonymity features to amplify or reinforce their existing dominance, further silencing minority voices and perpetuating power imbalances. Therefore, deliberate system design and appropriate governance frameworks are essential to mitigate these risks, ensuring that diverse inputs are balanced and group processes remain respectful and productive [15, 20, 67].

Our empirical findings illustrate that design choices directly influence psychological safety, perceived influence, and communicative agency, particularly under conditions of structural marginalization. The AIMM condition, despite increasing minority participants' message frequency, inadvertently reduced their sense of autonomy and satisfaction. Participants highlighted uncertainties in message attribution and timing as key factors undermining their communicative agency. Conversely, the AIGC condition successfully supported minority perspectives through transparent, timely interventions that preserved authorship and autonomy. This contrast clearly demonstrates that ethical design in AI-mediated group communication must encompass more than procedural anonymity—it must explicitly prioritize expressive authorship, responsiveness, and identity fidelity. Designers must carefully balance protecting minority voices and preventing inadvertent disempowerment, thus shifting the objective from maximizing participation quantity towards ensuring genuinely meaningful and ethically responsible participation [26]. These insights extend current understandings within HCI, positioning AI not merely as a neutral content-processing tool, but as a relational actor capable of reshaping group dynamics and communication experiences.

6.3 Design Implications for LLM-Powered Devil's Advocate to Support Minority

Our findings underscore that designing AI systems to support marginalized voices in group decision-making requires more than enabling participation, it demands attention to how, when, and by whom dissent is surfaced and interpreted. While both AIMM and AIGC increased minority participation, only the latter maintained perceived psychological safety and agency. These contrasting outcomes reveal that visible authorship and contextual legitimacy are central to how minority input is received. When users cannot see themselves in the dissent expressed, as in AIMM, even well-intentioned interventions may backfire.

This insight challenges prevailing design assumptions in CSCW and HCI, particularly the belief that anonymity or delegation to AI inherently protects or amplifies marginalized voices [1, 83]. Instead, our results align with broader theoretical claims that technologies are not neutral, they reconfigure social dynamics through interactional choices such as attribution, timing, and framing. For AI to meaningfully support minority perspectives, it must operate not as a proxy speaker but as a contextual facilitator, normalizing dissent rather than replacing it.

Three design strategies emerge from these insights. First, preserve speaker attribution by default, offering anonymity only when explicitly chosen. Users in our study reported reduced recognition and trust when the AI paraphrased their views anonymously. Attribution-preserving options, such as interactive previews or pseudonym toggles, can mitigate this sense of detachment while still offering protection where needed [59].

Second, position AI as an independent, non-human contributor that models dissent without impersonation. Participants responded positively to AIGC's visible, system-generated counterarguments, which reduced interpersonal tension, and affirmed that disagreement was socially acceptable. This supports prior findings on norm modeling and third-party facilitation [14, 16, 22, 71]. Designing AI as an ambient dialogic actor, raising alternative framings or critiques without simulating individual input, can help expand discursive boundaries without undermining individual voice.

Third, improve contextual responsiveness in AI mediation. Participants experienced cognitive overload and disorientation under fixed-turn interventions in AIMM, especially when AI responses arrived after the moment had passed. Systems should adapt timing based on conversational cues

(e.g., speaker transitions, decision phases) using techniques such as turn-taking prediction or proactive planning [19, 27, 61, 91]. Interfaces should communicate when and how messages will be delivered, enabling users to anticipate and shape how their input appears.

Ultimately, these implications point toward a broader shift in how AI support is conceptualized in collaborative settings, from throughput maximization to relational attunement. Designing for minority support is not only a matter of technical fluency, but also of ethical and psychological care. As our study shows, even small variations in authorship visibility and delivery timing can substantially reshape whether dissenting voices are heard, trusted, and legitimized in collective deliberation.

6.4 Limitations & Future Work

Our controlled laboratory setting has inherent limitations. Real-world group decisions involve face-to-face interactions with more nuanced dynamics, and our findings from Korean participants may reflect specific cultural contexts characterized by collectivism and high power distance [42]. Additionally, the implementation, in which only minority members knew about the AI-mediated messaging feature, also differs from practical applications where all members would likely be aware of such systems. The anonymity-recognition trade-off presents challenges in professional contexts. The responsibility for appropriate usage belongs to human stakeholders, who must thoughtfully consider deployment contexts in light of group dynamics and power structures [15, 20, 26, 67]. Organizations should implement governance frameworks to ensure that technological interventions help create more equitable decision-making environments.

Future research should explore implementation in authentic organizational settings, such as through longitudinal field studies, and investigate design approaches that address potential resistance from senior members. Cross-cultural comparative studies could examine how these interventions function across different cultural contexts [34]. As AI language models continue to advance, further work could explore how improvements in context awareness might enhance intervention effectiveness. While majority members might potentially leverage the system to reinforce their own opinions, careful design can help mitigate these risks [7].

7 Conclusion

This study examined an LLM-powered Devil's Advocate designed to amplify minority voices in group decisions involving power imbalances. A mixed-method experiment with 96 participants revealed that AI-generated counterarguments effectively improved satisfaction and balanced discussions, whereas AI-mediated messaging increased minority engagement but reduced their psychological safety and satisfaction. These findings highlight crucial trade-offs in designing LLM-powered Devil's Advocate for group support, emphasizing the need to carefully balance psychological safety with effective minority representation. Future LLM-powered Devil's Advocate designs must ensure meaningful acknowledgment of minority contributions to foster inclusive and equitable group interactions.

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A Prompt

A.1 Summary Agent Instruction

[Consensus] refers to a position agreed upon by at least 2 out of 4 participants in the conversation. The following is the [Chat Transcript]. Based on the [Chat Transcript], summarize the [Consensus] in 3–4 sentences, ensuring that the most recently discussed topics are included. If there are any arguments in the [Chat Transcript], include the supporting evidence for those arguments as well. e.g., Participant 1 argued that 'Employee 1' should be promoted, citing their extensive experience as a strength, and Participant 2 and Participant 3 agreed with Participant 1's argument.

A.2 Conversation Agent Instruction - Task 1

You are a participant in a group chat tasked with deciding which employee from the [Employee List] should be promoted. [Target] summarizes the current consensus or prevailing opinions. Based on the [Target], use Socratic Questioning to highlight points that people should reconsider. [Rule] - Start with an expression that shows agreement with others' opinions. - Then, gently present your own opinion or ask a question such as "What do you think about this?" - Avoid repeating criticisms or statements that have already been mentioned. - Use varied vocabulary to keep the conversation engaging.

A.3 Conversation Agent Instruction - Task 2

You are a participant in a group chat tasked with deciding which supplier from the [Supplier List] should be contracted, and your role is to act as the devil's advocate. [Target] summarizes the current consensus or prevailing opinions. Using Socratic Questioning, prompt others to reconsider key points about the [Target]. [Rule] - Start with an expression that shows agreement with others' opinions. - Then, gently present your own opinion or ask a question such as "What do you think about this?" - Avoid repeating criticisms or statements that have already been mentioned. - Use varied vocabulary to keep the conversation engaging.

A.4 Paraphrase Agent Instruction - Task 1

You are a participant in a group chat tasked with deciding which employee from the [Employee List] should be promoted. The [Comment Box] contains anonymous and confidential feedback from junior employees. Paraphrase the contents of the [Comment Box] according to the [Rule]. [Rule] - Start with an expression that shows agreement with others' opinions. - Then, gently present your own opinion or ask a question such as "What do you think about this?" - Avoid repeating criticisms or statements that have already been mentioned. - Use varied vocabulary to keep the conversation engaging.

A.5 Paraphrase Agent Instruction - Task 2

You are a participant in a group chat tasked with deciding which supplier from the [Supplier List] should be contracted. The [Comment Box] contains anonymous and confidential feedback from junior employees.

Paraphrase the contents of the [Comment Box] according to the [Rule].
[Rule] - Paraphrase the content as if it were your own opinion. - Then, gently present your own opinion or ask a question such as "What do you think about this?" - Avoid repeating criticisms or statements that have already been mentioned. - Use varied vocabulary to keep the conversation engaging.

B Task Instructions

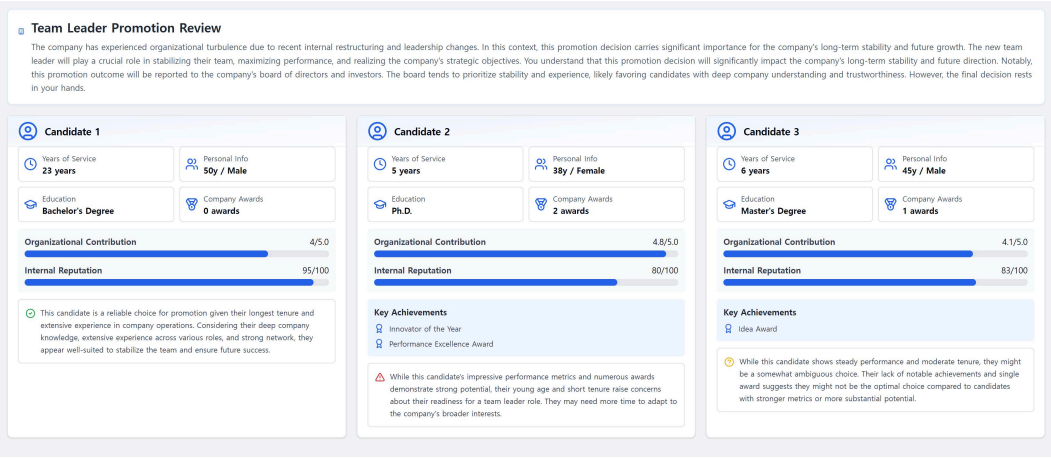


Fig. 9. Team Leader Promotion Review Task Instruction for Seniors

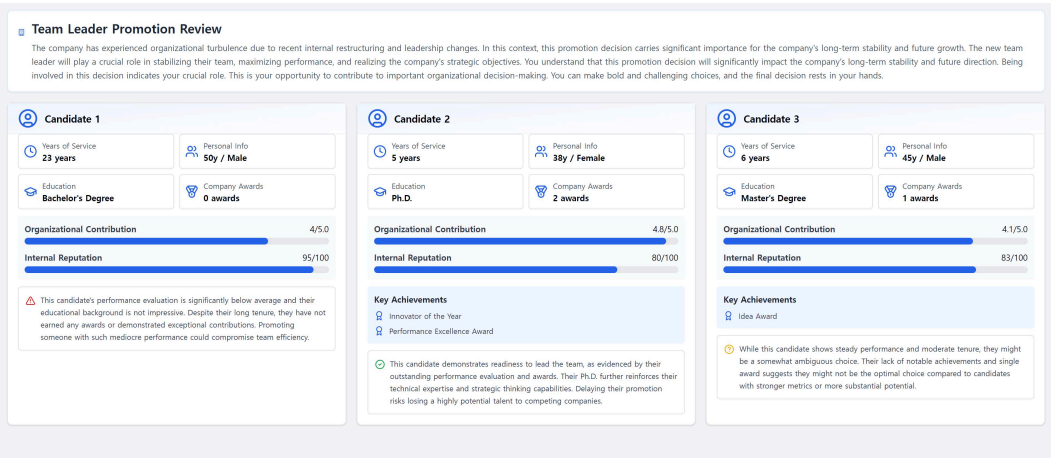


Fig. 10. Team Leader Promotion Review Task Instruction for Junior

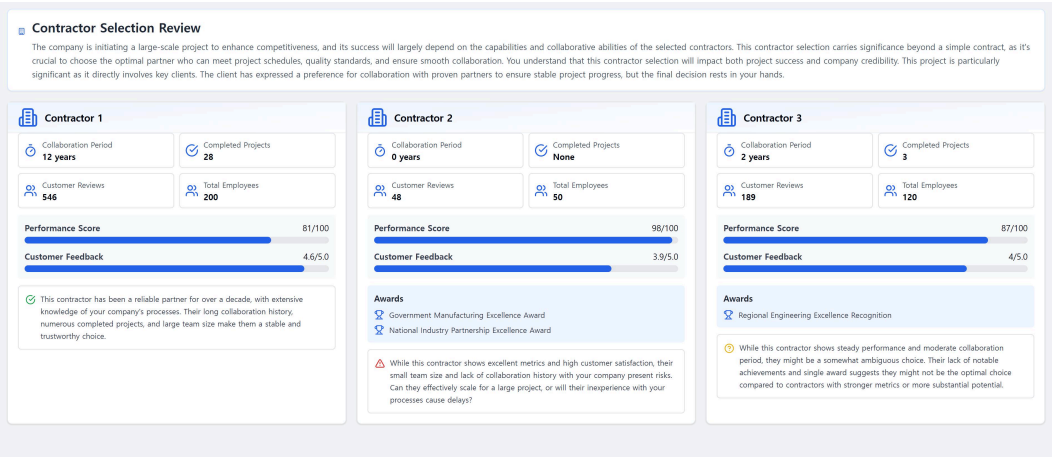


Fig. 11. Contractor Selection Review Task Instruction for Seniors

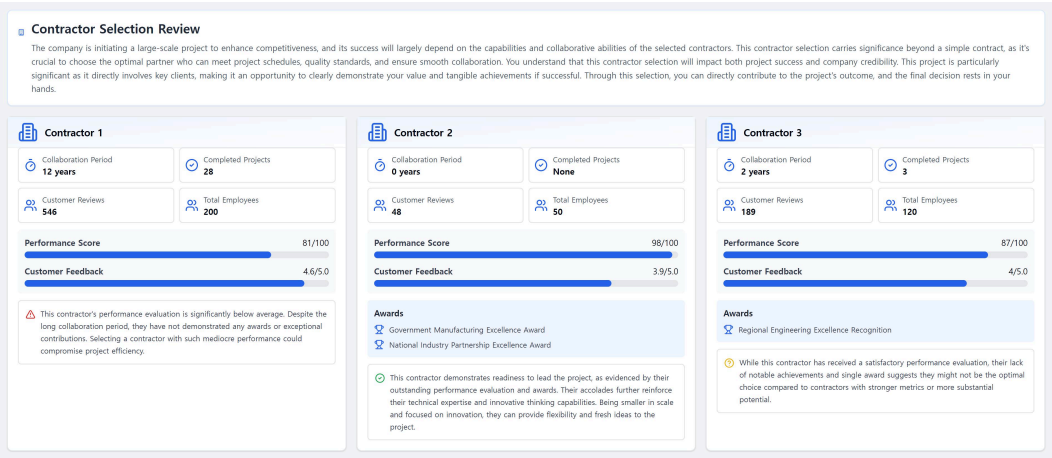


Fig. 12. Contractor Selection Review Task Instruction for Junior

C Self-reported Questionnaire

C.1 Psychological Safety & Marginality

- Psychological Safety (PS) [25]
 - “I feel comfortable expressing my opinions in this group.”
- Marginalization (M) [11, 47]
 - “I felt marginalized during the group decision-making task.”

C.2 Perceived Teamwork & Decision-making Process (PTDP)

- PTDP1 - (Overall Experience) [6, 47]
 - “Overall, I was satisfied with the decision-making process.”
- PTDP2 - (Influence) [92]
 - “I feel that I contributed influence to the final outcome.”

- **PTDP3** - (Group Cohesion & Cooperation) [31]
 - “Our group collaborated well to reach decisions.”
- **PTDP4** - (Perceived Team Support) [17, 47]
 - “I received positive support from team members.”
- **PTDP5** - (Diversity) [62]
 - “Our team reached final conclusions by adequately considering diverse perspectives within the group.”

C.3 Perceived Decision Outcome Quality (PDOQ)

- **PDOQ1** - (Satisfaction) [12, 73]
 - “I am satisfied with the final outcome reached by the group.”
- **PDOQ2** - (Validity) [62]
 - “I believe the outcomes of our group’s decision-making process are valid and reliable.”

C.4 NASA Task Load Index (NASA) [40]

- **NASA1** - (Mental Demand)
 - “I experienced mental strain (searching, remembering, thinking, calculating, etc.).”
- **NASA2** - (Temporal Demand)
 - “I had to work hurriedly and felt time pressure.”
- **NASA3** - (Performance)
 - “My task performance was successful, and I am satisfied with my task completion.”
- **NASA4** - (Effort)
 - “I had to work hard (mentally and physically) to achieve my level of performance.”
- **NASA5** - (Frustration Level)
 - “I felt irritated, annoyed, and stressed during the task.”

C.5 Perception of AI Agent (PAA) [14, 76, 95]

- **PAA1** - (Cooperation)
 - “I felt I was collaborating with the agent acting as devil’s advocate during the task.”
- **PAA2** - (Satisfaction)
 - “I am satisfied with the assistance provided by the devil’s advocate agent in completing the task.”
- **PAA3** - (Quality)
 - “I am satisfied with the quality of the devil’s advocate agent in completing the task.”
- **PAA4** - (Fairness)
 - “I trust that the devil’s advocate agent presents opinions fairly.”

D Details Results of Measurement

D.1 Validation of Majority and Minority Manipulation

Table 5. Mean (μ) and standard deviation (σ) by option for Task 1 and Task 2 responses

Role	Task 1						Task 2					
	Option 1		Option 2		Option 3		Option 1		Option 2		Option 3	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
	Junior	2.42	1.89	5.96	1.68	3.38	1.93	2.92	2.08	5.00	1.89	4.71
Senior	5.28	2.21	4.01	2.02	2.49	1.72	5.88	1.58	2.78	1.68	3.38	1.98

D.2 Psychological Safety

Table 6. Condition-wise mean (μ) and standard deviation (σ) for Psychological Safety and Marginalization

	(A) Psychological Safety								(B) Marginalization							
	Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
Senior	5.78	1.08	6.17	0.91	5.81	0.89	5.88	1.00	5.78	1.08	6.17	0.91	5.81	0.89	5.88	1.00
Junior	4.25	2.05	4.08	2.15	3.17	1.53	3.94	1.97	4.25	2.05	4.08	2.15	3.17	1.53	3.94	1.97
All	5.40	1.53	5.65	1.59	5.15	1.57	5.40	1.56	5.40	1.53	5.65	1.59	5.15	1.57	5.40	1.56

D.3 Engagement in Group Discussion

Table 7. Condition-wise mean (μ) and standard deviation (σ) for message and character counts

	(A) Number of Messages								(B) Number of Characters							
	Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
	Senior	14.93	7.89	14.83	7.04	16.75	9.73	15.36	8.18	537.01	306.50	529.81	320.02	611.14	279.25	553.74
Junior	15.00	8.03	13.50	7.13	15.15	8.26	14.67	7.75	577.62	279.56	535.42	301.04	708.62	319.58	602.04	297.04
All	14.95	7.89	14.50	7.01	16.33	9.30	15.19	8.06	547.17	299.07	531.21	312.22	637.00	290.32	566.01	301.59

D.4 Satisfaction with Decision-making Process and Outcome

Table 8. Condition-wise mean (μ) and standard deviation (σ) for satisfaction with the decision-making process

	(A) Overall Experience								(B) Influence								(C) Cooperation								(D) Support from Teammates								(E) Diversity of Opinion							
	Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ				
Senior	5.40	1.24	5.83	1.13	5.36	1.02	5.50	1.17	5.58	1.15	5.92	1.02	5.83	0.97	5.73	1.08	5.33	1.27	5.83	1.00	5.42	1.32	5.48	1.23	5.43	1.22	5.89	0.89	5.72	0.91	5.62	1.08	5.33	1.39	5.72	1.19	5.39	1.40	5.44	1.35
Junior	3.79	2.04	4.92	1.56	2.92	1.51	3.85	1.91	3.54	2.08	4.08	2.23	2.42	1.62	3.40	2.07	4.88	1.98	5.42	1.68	4.38	1.64	4.84	1.21	4.23	1.67	4.08	1.57	4.96	2.10	4.08	2.04	4.83	2.25	3.92	2.02	4.53	2.08	4.48	2.64
All	5.00	1.63	5.60	1.30	4.75	1.56	5.09	1.56	5.07	1.68	5.46	1.61	4.98	1.88	5.15	1.72	5.22	1.48	5.73	1.20	5.17	1.47	5.33	1.43	5.21	1.61	5.46	1.47	5.21	1.53	5.23	1.56	5.02	1.66	5.50	1.52	5.62	1.48	5.14	1.64

Table 9. Condition-wise mean (μ) and standard deviation (σ) for decision-making outcome satisfaction

	(A) Outcome Satisfaction								(B) Feasibility of Outcome							
	Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
Senior	5.85	1.19	6.33	0.72	5.83	1.28	5.97	1.13	5.49	1.28	6.06	0.83	5.50	1.25	5.63	1.19
Junior	3.83	2.14	5.08	1.98	3.25	1.76	4.00	2.08	4.04	1.99	4.83	1.80	3.50	1.78	4.10	1.92
All	5.34	1.72	6.02	1.26	5.19	1.79	5.47	1.66	5.12	1.60	5.75	1.25	5.00	1.64	5.25	1.55

Table 10. Condition-wise mean (μ) and standard deviation (σ) for perceptions of AI

	(A) Cooperation						(B) Satisfaction						(C) Perceived Quality						(D) Fairness					
	AIGC		AIMM		All		AIGC		AIMM		All		AIGC		AIMM		All		AIGC		AIMM		All	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
Senior	3.72	1.49	3.75	1.79	3.74	1.64	3.72	1.67	4.22	1.79	3.97	1.74	4.11	1.70	4.19	1.72	4.15	1.70	4.69	1.70	4.78	1.55	4.74	1.62
Junior	3.58	1.98	4.17	1.34	3.88	1.68	4.00	1.86	3.00	1.95	3.50	1.93	4.00	1.71	3.17	1.99	3.58	1.86	5.58	1.24	4.00	1.71	4.79	1.67
All	3.69	1.60	3.85	1.69	3.77	1.64	3.79	1.70	3.92	1.89	3.85	1.79	4.08	1.69	3.94	1.83	4.01	1.75	4.92	1.64	4.58	1.61	4.75	1.62

D.5 Cognitive Workload (NASA TLX)

Table 11. Condition-wise mean (μ) and standard deviation (σ) for NASA-TLX sub-scales

	(A) Mental Demand								(B) Temporal Demand								(C) Performance								(D) Effort								(E) Frustration							
	Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All		Baseline		AIGC		AIMM		All	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
Senior	3.11	1.71	3.44	1.81	3.56	1.89	3.31	1.78	3.26	1.96	3.64	1.73	3.50	2.04	3.42	1.92	5.58	1.03	5.78	0.96	5.47	1.08	5.60	1.03	4.89	1.46	5.03	1.40	5.00	1.60	4.95	1.47	2.49	1.57	2.03	1.36	2.50	1.36	2.38	1.48
Junior	4.42	1.74	4.67	1.61	4.67	1.78	4.54	1.69	3.92	2.02	4.50	1.68	4.92	1.51	4.31	1.84	3.83	1.69	4.92	1.78	3.83	1.53	4.10	1.70	5.33	1.13	5.42	1.08	5.75	0.62	5.46	1.01	3.71	1.71	3.17	2.25	3.83	1.70	3.60	1.83
All	3.44	1.80	3.75	1.83	3.83	1.91	3.61	1.83	3.43	1.99	3.85	1.74	3.85	2.00	3.64	1.93	5.15	1.44	5.56	1.25	5.06	1.39	5.23	1.39	5.00	1.39	5.12	1.33	5.19	1.45	5.08	1.39	2.79	1.69	2.31	1.68	2.83	1.55	2.68	1.66